BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

File Handling

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is the scope of a local variable or constant?

2. What happens in the fetch stage?

A) Get the next instruction
B) Work out its meaning
C) Carry out the instruction
D) Save a backup copy

3. "Execution only copies an instruction into memory without performing it." Correct this. Explain why.

4. "The Control Unit stores your documents forever." Correct this. Explain its role when the CPU runs an instruction.

Today's objectives

Explain the core ideas of file handling using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Open
A file is opened before reading or writing through a program.

Read
Input from an existing file can be processed by a program.

Write
Output can be saved to a file, with appropriate mode and permissions.

Close
Closing releases the file resource and helps ensure changes are saved.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Persistent storage	A text file keeps data after the program ends. A filename identifies a file, but does not itself hold the file's contents.
Open and read	OCR ERL stores an opened file in a variable: f = open('pupils.csv'); f.readLine() returns the next line. This file has a header.
Write and close	f.writeLine(text) writes a line to an opened output file; f.close() releases the file handle.
CSV and fields	In this teaching example, each line separates ID, Name and Mark with commas. Splitting CSV fields into an array is an optional extension, not a separate compulsory OCR requirement.

I Do - One worked example

The sample file pupils.csv contains a header followed by 101,Ada,72; 102,Ben,43; 103,Mia,81. Trace two readLine() calls and state the lines returned.

1. Open pupils.csv with file = open('pupils.csv'); save the opened file in a variable.

2. First readLine() returns the header: ID,Name,Mark. The next readLine() returns 101,Ada,72.

3. Each further readLine() advances to the next line. Use endOfFile() when reading the whole file.

4. Close the file. Reading a line returns text; it does not automatically turn comma-separated fields into a 2D array.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A program creates a separate results.txt containing the lines 'Ada,72' and 'Mia,81'. Outline how it writes and then closes the file. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the instruction that prints each successive line in the supplied file.

```
myFile = open("sizes.txt")
while NOT myFile.endOfFile()
print(...)
endwhile
myFile.close()
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Name the four basic file-handling operations required by OCR J277. [4 marks]

Q3. Given pupils.csv with a header followed by 101,Ada,72 and 102,Ben,43, state what the second call to readLine() returns after opening the file. [2 marks]

Q4. Explain why the file should be closed after reading its final line. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A file should normally be closed after use. ☐ **T** ☐ **F**
2. A text file keeps data after the program ends ☐ **T** ☐ **F**
3. OCR ERL stores an opened file in a variable: f = open('pupils.csv'); f.readLine() returns the next line ☐ **T** ☐ **F**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 A local variable can be used only within its own subprogram or local scope. ☐

Q2 A - Get the next instruction ☐

Q3 The operation specified by the decoded instruction is carried out. ☐

Q4 The Control Unit decodes instructions and controls CPU operations. The ALU processes data; registers store it briefly. ☐

Code First - Model

```
myFile = open("sizes.txt")
while NOT myFile.endOfFile()
print(myFile.readLine())
endwhile
myFile.close()
```

Output: 4
6
10 (if sizes.txt contains these three lines)

We Do - Check the model

1. Close the file when finished; identify the order open -> write -> write -> close. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Open [1], Read [1], Write [1], Close [1]. ☐

Q3 101,Ada,72 [1]; because the first readLine() already returned the header [1]. ☐

Q4 Releases the opened file resource/handle [1] and properly completes the file access [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: